

Team121MLProject [GitHub Pages](#)

Road Condition Forecasting – Final Report

Final deliverable for CS 4641: Road surface condition forecasting

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Introduction/Background

Cities often repave roads only after a backlog of requests grows, and the worst conditioned roads end up being prioritized first. This delay means a street can wait months or even years before maintenance occurs. By the time crews arrive, the road is further deteriorated, which drives up costs and increases crash risk. Looking into San Francisco's data and approach: the Pavement Condition Index (PCI) rates each segment on a 0–100 scale, where 0 is a severely distressed roadway and 100 is a newly paved surface [2], [3]. PCI comes from MTC surveys and feeds into the StreetSaver platform, which forecasts future PCI using factors such as pavement age, climate and precipitation, traffic volume, available maintenance funding, street type (pedestrian, transit, bike, or cars/trucks), land use, and conflicts with other projects [5], [4].

Public Works publishes a rolling 5-year repaving plan and enforces a 5-year excavation moratorium after resurfacing [1]. Utilities are incentivized to coordinate work before paving or pay significant fees if they dig within the moratorium. Our project aims to recreate a PCI-like forecast so cities can schedule earlier, lower-cost interventions. Datasets include SF street-level PCI scores and federal traffic/climate resources [6]–[8].

Problem Definition

Problem: Pavement quality detection lags behind optimal repair schedules, and maintenance requests are often late, so pavement continues to degrade while waiting in the queue.

Motivation: This model aims to predict road maintenance needs early to make streets safer for pedestrians, drivers, and minimize lifecycle costs so cities can better allocate funds. We aim to predict near-term PCI (Pavement Condition Index) and flag pavement segments that are at risk to align with the 5-year plan (public works must publish a 5 year repaving plan: once a road is paved, nothing on that road can be touched “5 year excavation moratorium”) and utility coordination.

Methods

Data preprocessing methods

- Encoded True/False fields to binary (1/0) across datasets.
- Dropped rows with excessive null values and standardized null types.
- Removed irrelevant columns before merging data.
- Merged shoulder, drainage, IRI, traffic, and unbound_layers datasets via left joins on keys such as STATE_CODE, SHRP_ID, CONSTRUCTION_NO, YEAR.
- Removed duplicate rows/columns.
- Ran PCA to reduce 59 features to 39 while retaining roughly 95% of the variance to cut noise and training time.
- **Feature engineering:**
 - **Pavement Age:** Calculated as years since construction; older segments deteriorate faster and accumulate roughness more quickly.
 - **Survey Interval:** Captures gaps between inspections; large intervals can hide accelerated deterioration.
 - **Lag MRI:** Previous inspection year's MRI; a strong predictor of next-year MRI that captures historical deterioration trends.
 - **MRI Delta:** Change in roughness between consecutive surveys; helps the model learn segment-specific deterioration rates.

Algorithms and models (3+)

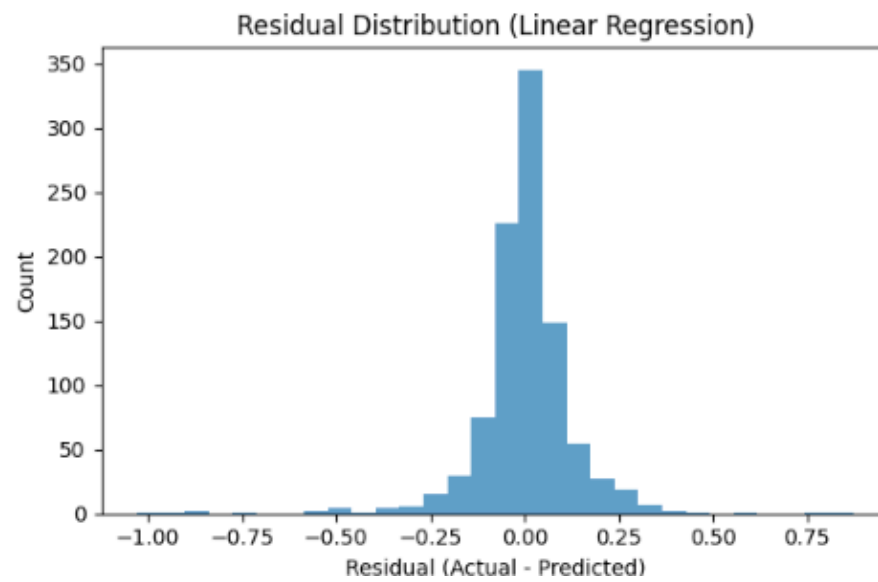
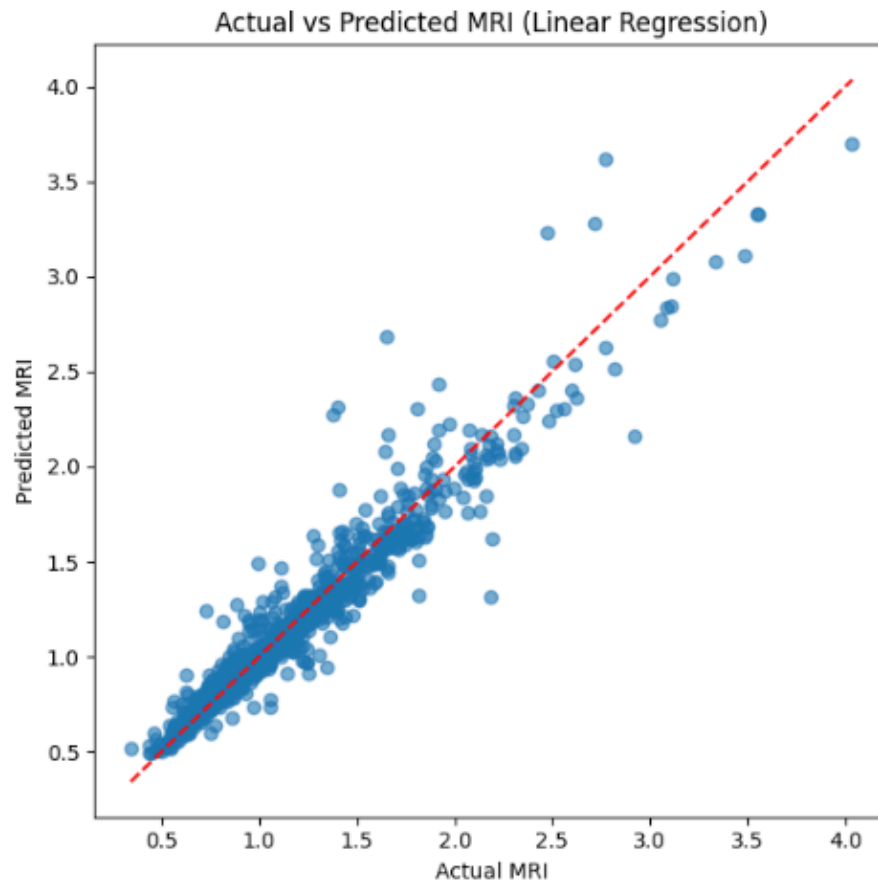
- **Baseline Linear Regression:** Full feature set to establish a reference fit for MRI_next.

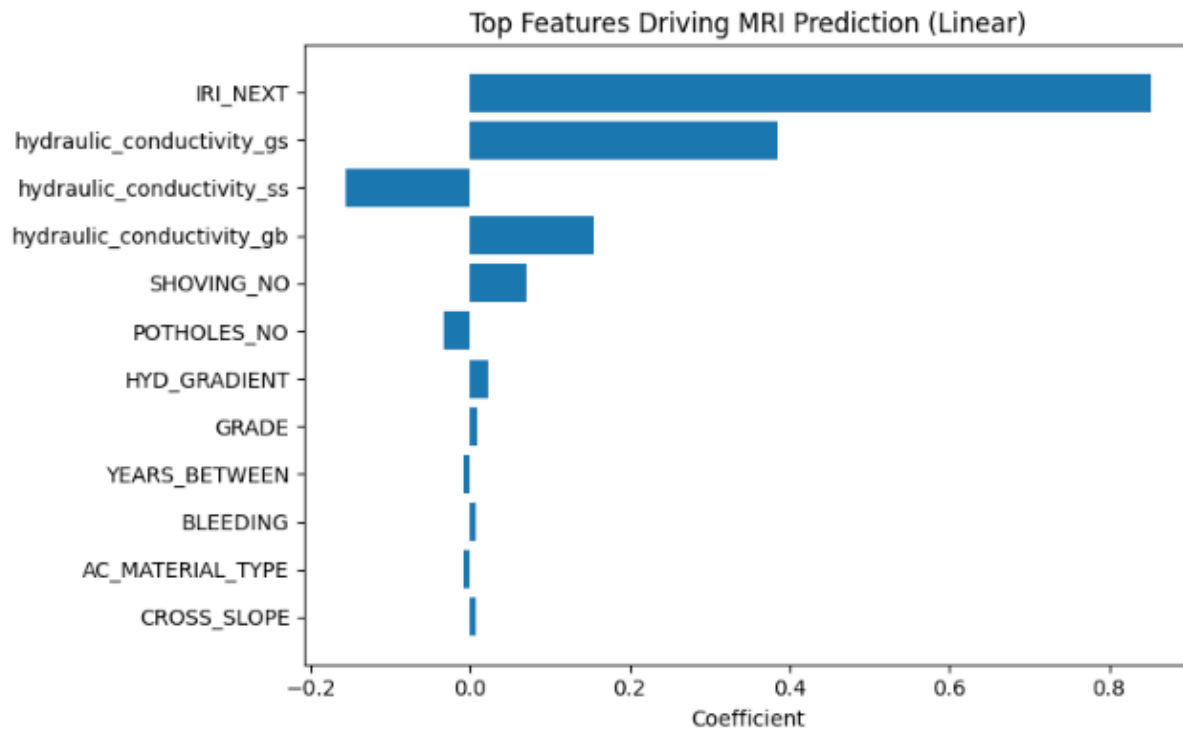
- **Linear Regression with PCA:** Dimensionality reduction to focus on the top components while shortening training time and reducing noise.
- **Ridge Regression:** Distributes weight across correlated features to stabilize coefficients.
- **Lasso Regression:** Drives unimportant coefficients to zero to simplify the model.
- **Random Forest Regressor:** Captures non-linear relationships and feature interactions through bagged decision trees.
- **Gradient Boosting Regressor:** Sequentially corrects residual errors to improve forecasts on difficult segments.

Multiple linear variants help interpret coefficients and identify signal strength, while the tree ensembles (Random Forest and Gradient Boosting) explore non-linear effects and feature interactions to improve predictive performance on roughness hot spots.

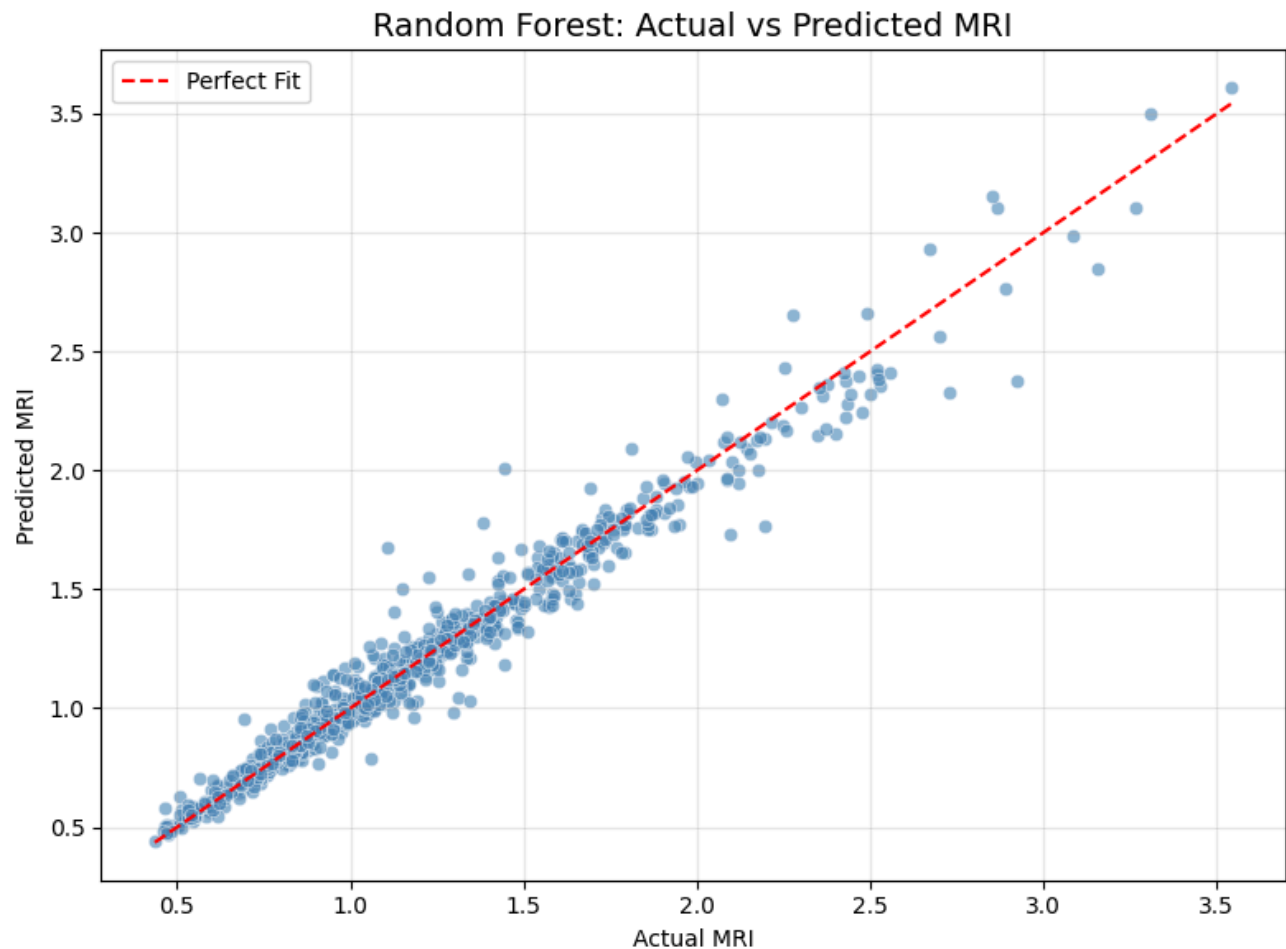
Results and Discussion

Visualizations

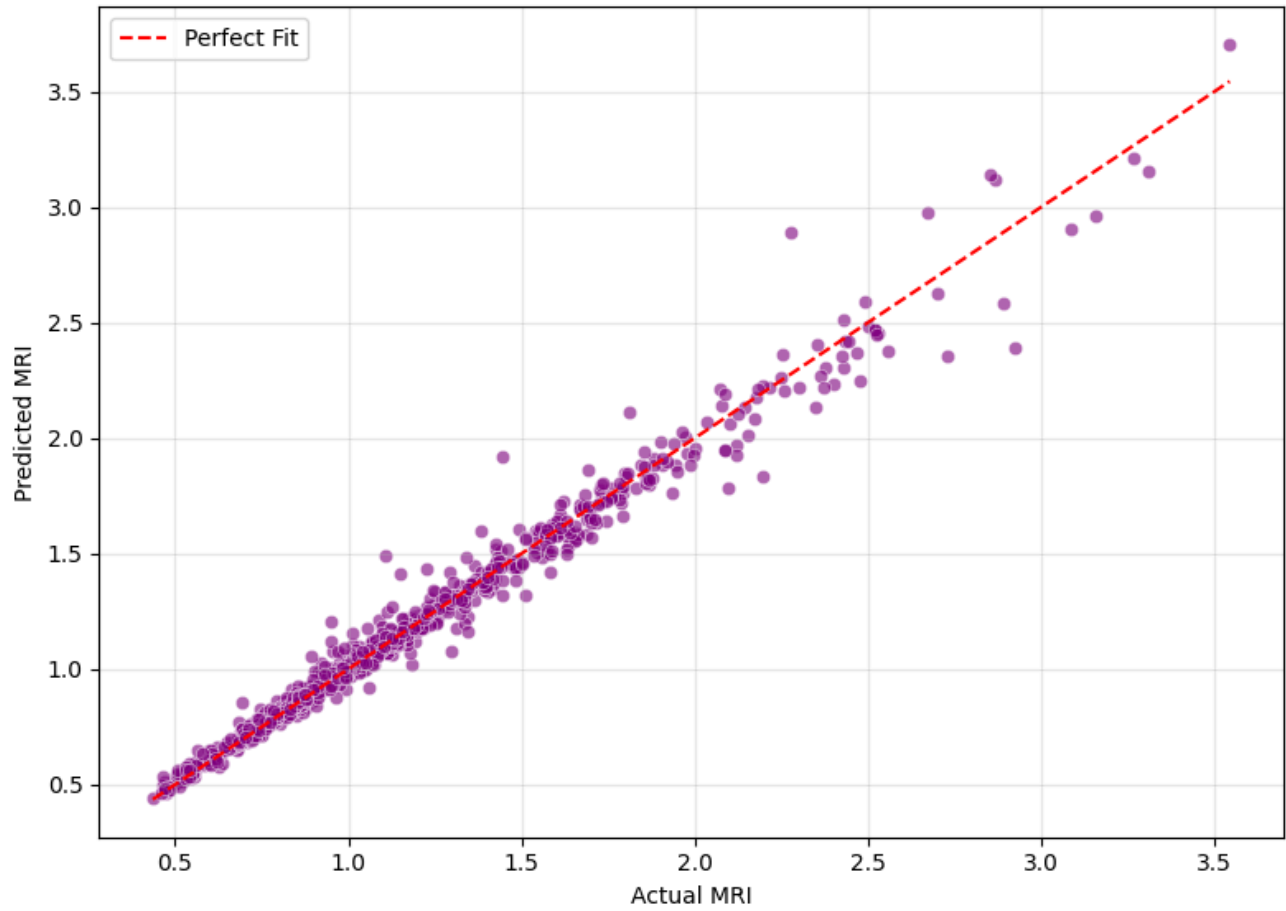


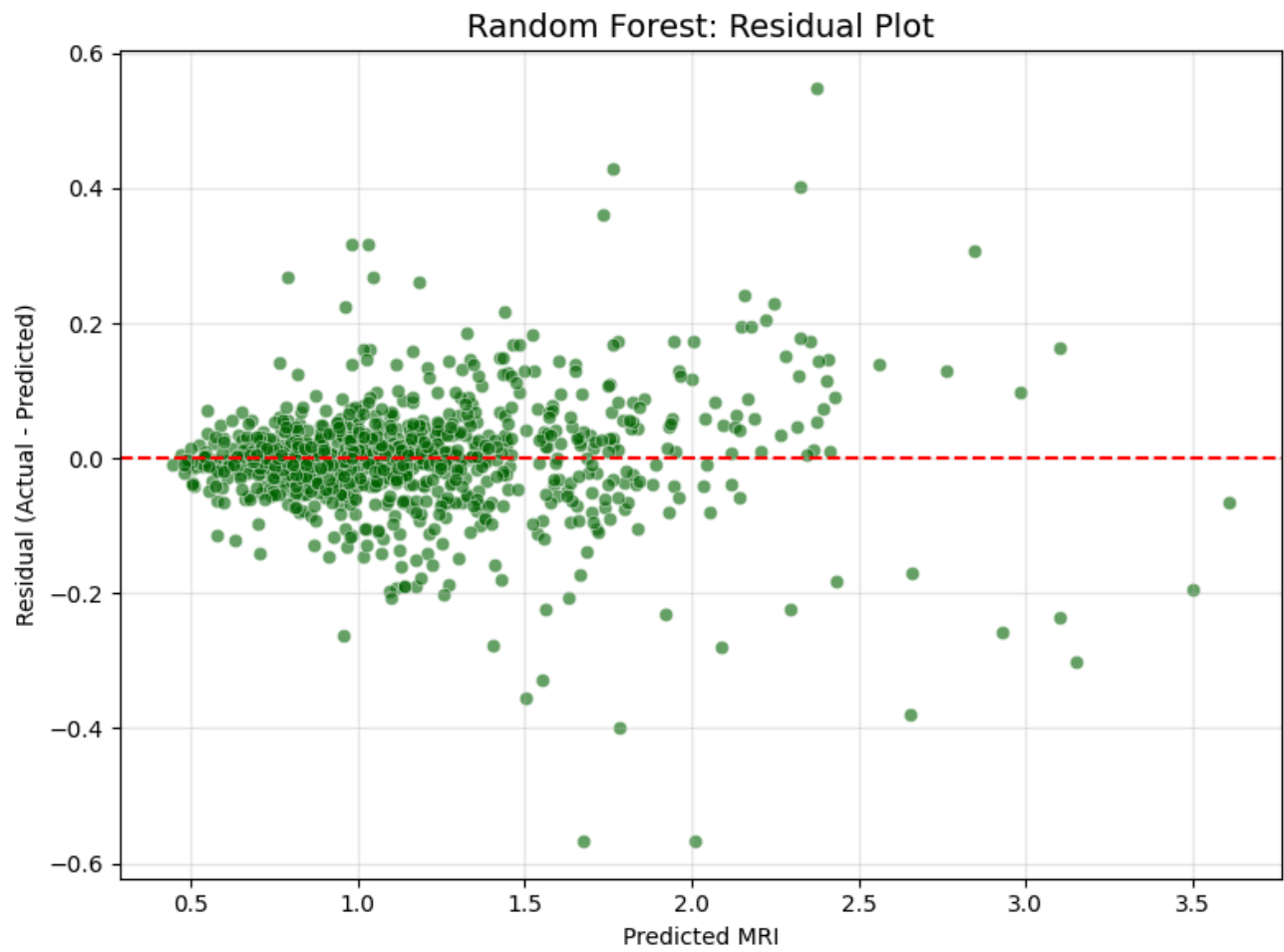


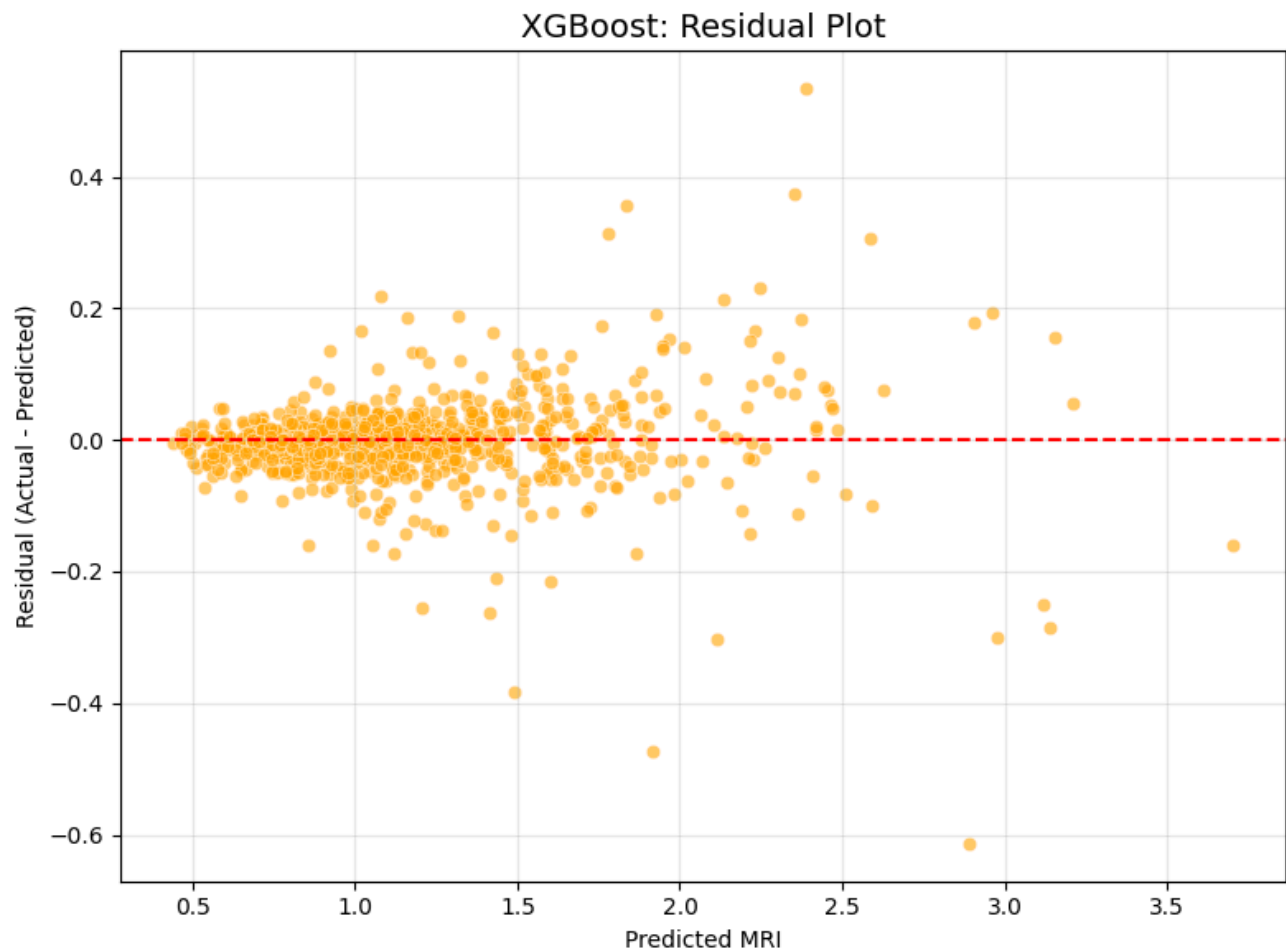
Model comparison visuals



XGBoost: Actual vs Predicted MRI







Quantitative Metrics

- **R²:** How well the regression line predicts the outcome of a dataset, measured as the proportion of variance in the output that is captured by the features of the model.
- **RMSE (Root Mean Squared Error):** Square root of average of squared deviations; penalizes larger residuals and converts back to original units.

Model comparison metrics (3+ models)

Model	R ²	RMSE	Notes
Baseline Linear Regression	0.922	0.134	Reference fit; fully interpretable coefficients.
Random Forest Regressor	0.9678	0.0839	Captures non-linear effects and interactions; monitor overfitting.
XGBoost Regressor	0.9802	0.0657	Boosted trees capture complex effects; strongest performance observed.

Analysis and comparisons

We split the data into training (80%) and testing (20%) sets. After iterative feature cleanup (dropping SHRP_ID/state codes, standardizing nulls, encoding PAVEMENT_FAMILY), the baseline linear model reached $R^2 = 0.924$ and $RMSE = 0.138$, showing a solid interpretable fit.

Tree ensembles vs. linear: Random Forest ($R^2 = 0.9678$, $RMSE = 0.0839$) and XGBoost ($R^2 = 0.9802$, $RMSE = 0.0657$) substantially improve accuracy. In the actual-vs-predicted plots, XGBoost points cluster closest to the perfect-fit line, especially for rougher segments ($MRI > 2.5$). Random Forest remains strong but shows slightly wider spread at the high end.

Residual behavior: Residual plots center near zero for both ensembles. XGBoost residuals show tighter variance across the range; Random Forest residuals widen for mid-to-high MRI, indicating higher variance. Neither model exhibits pronounced bias, but XGBoost generalizes better on the roughest segments.

Model choice and tradeoffs: Linear regression is the most interpretable baseline. XGBoost currently provides the best predictive performance and smoother residuals; Random Forest is a close second and may be preferred if training speed or interpretability (via feature importances) is prioritized. Ridge/Lasso and PCA-linear remain available for coefficient stability and dimensionality reduction when interpretability is paramount.

At the end of the analysis, we examined the weights of all the features as a sanity check to ensure that the model's predictions made sense. The weights were consistent with our expectations: for example, the Mean Roughness Index (MRI), which represents the average of the left and right wheel path IRI values, had a weight of approximately 1. This indicates that MRI is a very important feature, having a high impact on predicting MRI_{next} , as it directly reflects the road's roughness. On the other hand, most of the other features had weights close to 0.01, which suggests that they have a smaller influence on the outcome in comparison to MRI.

Next Steps

Currently, the model does a great job in predicting the outcomes of pavement condition. However, once again the model bases its decisions on a select few features, as a result when continuing this project we would like to try to find other determining factors when it comes to pavement condition. In addition, since our algorithms have been proven to be quite successful, if we are able to find an algorithm that is able to nearly perfect its accuracy, we would love to have some sort of implementation in the real world. This implementation would aim to notify those in charge of paving the roads preemptively so that we can ultimately accomplish our goal of reducing these road related accidents.

Gantt Chart

This final Gantt chart captures planned responsibilities across the full project timeline.

Final Contribution Table

Member	Final Contribution
Akshar	Built linear models and performed PCA, set up GitHub pages, worked on writing report.
Manthan	Evaluated model performance, created slides for performance metrics, wrote report content.
James	Built Random Forest and Boosting models and visuals to depict each model's forecasts, worked on writing report, updated the GitHub README.
Cyrus	Wrote large portion of report and worked on building final presentation, worked on writing report.
Vidul	Created slides for hyperparameters and why model, worked on writing report and github pages.

References

- [1] "Street Resurfacing Program | Public Works," Sfpublicworks.org, Oct. 2024. Available: <https://sfpublicworks.org/street-resurfacing>. (accessed Oct. 04, 2025).
- [2] "Pavement condition index (PCI) | SF.gov," Sf.gov, 2024. Available: <https://www.sf.gov/data--pavement-condition-index-pci>. (accessed Oct. 04, 2025).
- [3] "Pavement Conditions Index (PCI) | Metropolitan Transportation Commission," Mtc.ca.gov, Mar. 17, 2021. Available: <https://mtc.ca.gov/.../pavement-condition-index>.
- [4] "Street Pavement Condition - Vital Signs - SF Bay Area," Ca.gov, 2025. Available: <https://vitalsigns.mtc.ca.gov/.../street-pavement-condition>. (accessed Oct. 04, 2025).

5. [5] "StreetSaver® | Metropolitan Transportation Commission," Mtc.ca.gov, May 15, 2021. Available: <https://mtc.ca.gov/.../streetsaverr>.
6. [6] F. Public, "Streets Data - Pavement Condition Index (PCI) Scores," Sfgov.org, Sep. 25, 2012. Available: <https://data.sfgov.org/.../PCI-Scores>. (accessed Oct. 04, 2025).
7. [7] "Office of Highway Policy Information - Policy | Federal Highway Administration," Dot.gov, 2017. Available: <https://www.fhwa.dot.gov/.../tvvt.cfm>.
8. [8] "LTPP InfoPave - Analysis Ready Datasets," Dot.gov, 2017. Available: <https://infopave.fhwa.dot.gov/Data/ReadyMaterial>. (accessed Oct. 04, 2025).